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SMTCNN - A global spatio-temporal texture convolutional neural network for 3D dynamic texture recognition

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ABSTRACT

Dynamic textures (DT) are typically 3D videos of physical processes showing statistical regularity but have indeterminate spatial and temporal extent. Existing DT recognition methods usually neglect the global spatio-temporal relationships of DT which reflect the statistical regularities. In this paper, a spatio-temporal texture convolutional neural network (SMTCNN) is proposed for global semantic DT representation. Specifically, SMTCNN describes DT features by learning DT' temporal motion as well as the sources of the motions and the scenarios where the motion is happening, and accordingly, a motion net and a source net are formulated. In particular, a novel module consisting of expansion and concatenation implementations on deep features is presented, with an arbitrary 2D backbone as input, followed by a new 1D CNN including 4 convolutional, 2 pooling and 2 fully-connected layers to represent the 2D tensors in space–time, by transforming DT descriptors from discrete "words" to global "textures". A number of comparative experiments on three DT dataset - UCLA, DynTex and DynTex++ are conducted to demonstrate our approach.

1. Introduction

Human-robot interaction is enriched if the perception of the environment by the latter is accurate [1–3]. More meaningful interactions can result in such cases. However, the working environments of robots are mostly dynamic, with some temporal regularity. To this end, the topic of dynamic texture recognition is derived. Dynamic textures (DT) are time-varying 3D sequences of moving scenes that exhibit certain temporal statistics or stationary properties in time. Different with the other two visual motion signals-activities and motion events, DT are typically videos of physical processes showing statistical regularity but have indeterminate spatial and temporal extent [4], e.g. fire, smoke, sea-waves, foliage etc. Compared to static, or spatial texture analysis, the fuzziness of spatio-temporal extent of DT is greater [5] since in addition to the unlimited degree of freedom of spatial extent, it rather relates to their relationships, which is not just an addition of 2D spatial information-appearance. The variation of spatial extent with time is complex, however, this change is proved to be more important in action recognition tasks [6–8]. Considering the temporal change - the motions of DT are less distinct, e.g. smoke can be partially transparent. Also, since the categories between DT are more general and fuzzy (e.g. gentle sea waves and rough turbulent water), recognizing DT is challenging and attracting growing interest [9–11].

DT features can be roughly categorized into hand-crafted and learned. The former involves optical flow based methods [12,13] owing to the fact that not only the estimation of optical flow is computationally efficient, but also optical flow helps to transform the DT representation problem to concatenation of a sequence of instantaneous motion patterns. In spite of optical flow being effective for characterizing the local dynamics of DT, it is also adopted as a complementary stream [6] in convolutional neural network (CNN) based approaches detailed in the following section; however, the global temporal relationships of the motion are disregarded. Using spatio-temporal filtering is another popular hand-crafted technique to represent DT. Following 2D filters, a variety of 3D filters (e.g. 3D Gaussian [14] and 3D (Local Binary Patterns) LBP [15]) are proposed to extract the spatio-temporal features of

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DT. As the spatio-temporal filtered features are usually large, a statistics algorithm such as bag of words (BoG) is often employed for a more compact representation.

As to the learned DT features, early researches rely strongly on Linear Dynamical Systems (LDS), and the majority downside of those methods are the inability to represent patterns beyond the training data. Large sized datasets have driven progress in computer vision over the past few years. Unfortunately, LDS based learning as well as hand-crafted techniques are not suited for training on large volumed DT datasets. So, more focus on DT recognition is turned to CNN recently. There are two broad approaches to realizing spatio-temporal CNN. One is to characterize spatio-temporal features based on 2D CNN. Due to 2D CNN works only on space domain, further processing is needed, such as concatenation, combination of recurrent neural network (RNN) etc. The other way is to implement a 3D CNN, which is able to capture the global temporal relationships of sequences, however, the relatively small size of the existing dataset and the configuration of the 3D CNN architecture are the two points to be concerned with.

In this paper, we present an efficient SMTCNN for learning DT by considering the motions happened in DT as well as the sources of the motions and the scenarios where the motion is happening. Accordingly, a motion net and a source net are formulated. Specifically, the motion net first spatially extracts 2D CNN deep features of video frames, followed by further temporally representing spatial features using CNN "word", "text" and "abstract" progressively and semantically. Each word corresponds to a set of feature maps (7×7) of each frame from the last pooling layer of ResNet [16] and is composed of m (the number of feature maps) alphabets generated by expanding each feature map. To construct the spatio-temporal CNN text for the videos, the words ($m \times 49$) of all the frames (n) are concatenated. Therefore, the CNN text consists of m CNN sentences ($n \times 49$). To perceive the CNN text, 1D convolution is conducted to extract the abstract spatio-temporal features, and the abstract features are mapped to a fully connected layer. Softmax is employed for the final classification, and the overview is shown in Fig. 1. Since the implementations (expansion, concatenation and a 1D CNN) within our SMTCNN are simple and there is no extra parameter learning, SMTCNN is efficient. Moreover, the implementations also guarantee that the spatial features of each frame are adequately captured while the temporal information is taken into account to a large extent. As to the source net, because of the stationary characteristics of DT, to recognize the subjects more precisely, transfer learning based on ResNet and spatio-temporal slicing based data augmentation schemes are adopted for classification. The major contributions of this paper can be summarized as follows:

1. A novel deep learning architecture named SMTCNN is proposed for spatio-temporal semantic representation by considering DT' motion as well as the sources of the motions and the scenarios where the motion is happening.
2. A module consisting of tensor expansion, concatenation and a 1D CNN embedded in SMTCNN is presented to effectively learn the global spatio-temporal relationships of 2D CNN tensors.
3. We get the state-of-the-art DT recognition results by conducting a number of comparative experiments on three datasets.

We present the related work in Section 2, followed by section 3 that introduces our method, experiments are shown in section 4, while section 5 concludes the paper.

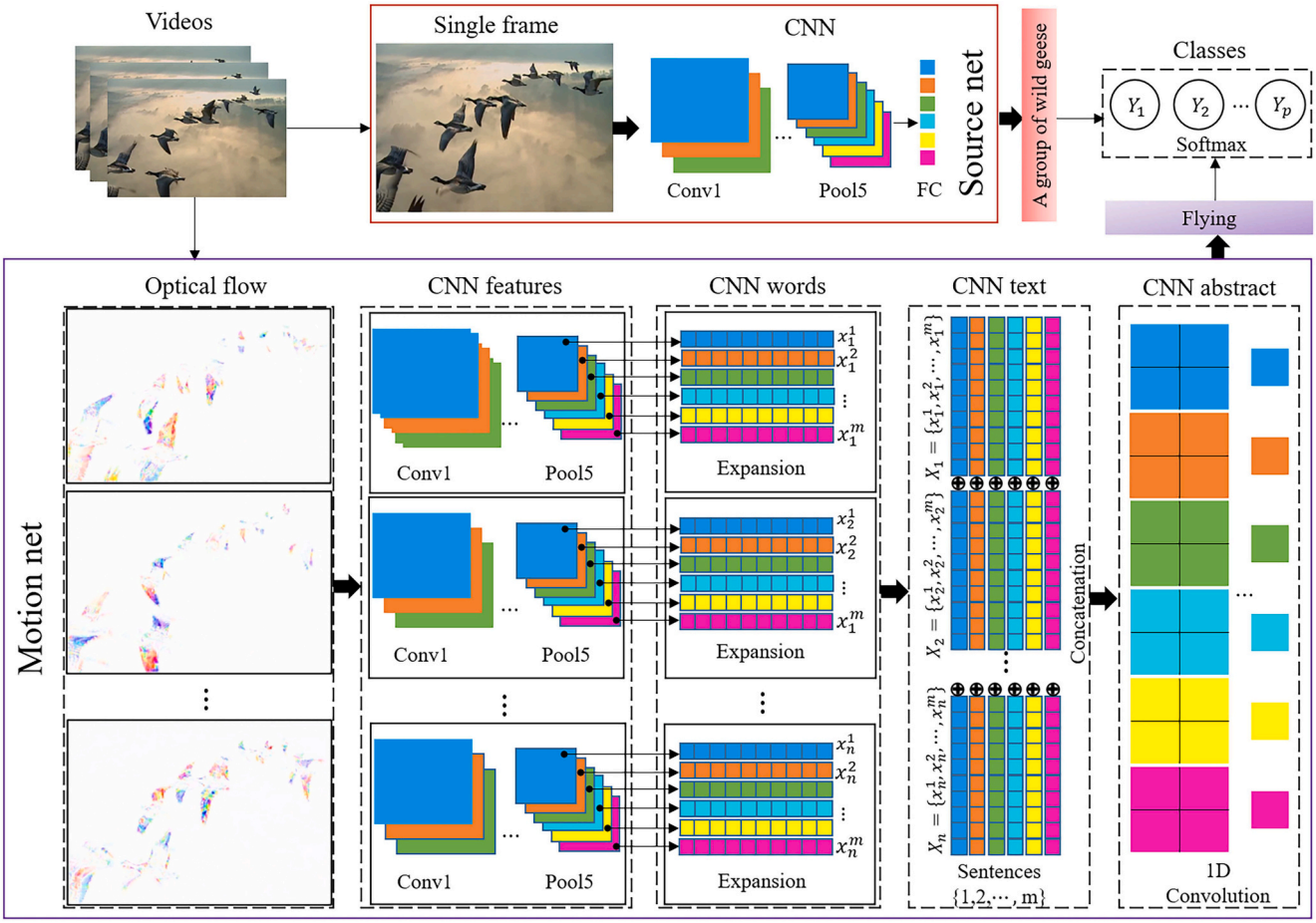


Fig. 1. Schematic of the proposed SMTCNN for DT recognition.

2. Related work

Similar to other visual recognition tasks [17–19], DT recognition approaches can be classified into hand-crafted feature based methods and deep learning methods.

Local binary pattern (LBP) feature is widely used in DT recognition since its strong ability to extract local gradient features, which is suited to handle the complex DT scenes with scale, illumination and viewpoint variation. In [20], LBP is first extended to volume LBP by considering the relationships between pixels and their neighbors, followed by a bit-wise transition encoding scheme to remove the rotation invariance of LBP, and finally extracted on three orthogonal planes (LBP-TOP - *xy*, *xt* and *yt*) to represent the spatio-temporal DT features. Tiwari and Tyagi [21] continue the LBP-TOP work and build a structure pattern descriptor by thresholding the LBP using Weber's law and multi-resolution analysis for DT recognition. Also, inspired by LBP-TOP, Zhao et al. [22] present CVLBC as the representation for both appearance and motion features of DT, by only counting the number of 1 s in the thresholded codes. Moreover, based on LBP, Nguyen et al. propose two effective DT descriptors - MMDP-B and MLDP-TOP by taking into account the advantages of filtering and local features for DT recognition in [23], and they introduce another two DT descriptors - FDT and FD-MAP by exploiting local features and chaotic motions along beams of dense trajectories in [24]. Quan et al. recognize DT by presenting STLS [25] to encode the LBP features using lacunarity analysis, and they continue to deal with the DT analysis problem by proposing two dictionary encoding and learning algorithms - Orthogonal Tensor DL and Equiangular Kernel DL in [26,27], respectively. Particularly, DFS, which consists of a volumetric fractal spectrum component (V-DFS) and a multi-slice fractal spectrum component (S-DFS) is studied in [28] for DT analysis. DNG, a spatio-temporal directional number transitional graph is proposed in [29] to replace the LBP features for DT recognition.

DT recognition problem is increasingly studied by CNN recent years. For instance, Qi et al. [30] apply a 5-layer CNN to extract DT features, followed by a statistics algorithm for recognition. In [13], two classic CNN architecture - AlexNet and GoogleNet are separately employed on three orthogonal planes for DT classification. Similarly, Hong et al. [31] adopt pre-trained CNN for the DT recognition issue by proposing D3, which uses key spatio-temporal segments of videos as the input of CNN. DT recognition is a typical spatio-temporal analysis issue, which can be well solved by revisiting and borrowing some recent work on CNN based video spatio-temporal representation. As far as we know, the initial work to extend spatial CNN to spatio-temporal CNN dates back to [32], where a set of hardwired kernels are first employed to generate dozens of feature maps of the action sequences, followed by three 3D convolution and two sub-sampling and one fully connected layers. In [33], to represent the action videos spatially and temporally, Yao et al. extract three hand-crafted features (HOG, HOF, motion boundary) of the video clips as the input of a 3D CNN with three convolution and pooling layers. Also, a temporal attention mechanism is proposed based on LSTM for automatically selecting the most relevant temporal segments. A Tube CNN consisting of five 3D convolution and four pooling layers is proposed in [34] for action detection, by linking the tube proposals of different video sequences using a scoring scheme named network flow. Tran et al. [35] introduce C3D for learning video spatio-temporal features and prove that small $3 \times 3 \times 3$ convolution kernels in all layers performs best for 3D CNN. Hadji et al. [9] presents a hierarchical spatio-temporal orientation representation for DT recognition, using 3D convolution and a novel repeated filtering technique. The repeated filtering is achieved by recurrent connection, and the recurrent nature makes the output of each layer of processing feeds back to the input. In [10], ResNet [16] is extended to ST-ResNet for DT recognition.

Beyond 3D convolution, some researchers handle the spatio-temporal learning problem by combining 2D CNN and recurrent neural network (RNN), such as [7,36,37]. Moreover, Simonyan and Zisserman [6] propose two-stream CNN to characterize spatio-temporal features using image frames and optical flow color maps as inputs, by considering both appearance and motion information of image sequences at the beginning of CNN, two-stream CNN dominates the action recognition [6] benefiting from the good generalization of the existing 2D CNN architecture. Sun et al. [38] propose a factorized spatio-temporal CNN that factorizes the original 3D convolution kernel learning as a sequential process of learning 2D spatial kernels in the lower layers (called spatial convolutional layers), followed by learning 1D temporal kernels in the upper layers (called temporal convolutional layers). Recently, some researchers try to deal with the DT representation problem using attention mechanism based visual transformers [39,40], however, the trade-off between computation cost, data volume for training and accuracy is not that promising. Plus, existing methods neglect the spatio-temporal relationships of DT that reflect the statistical regularities, leading to poor accuracy for cases with complex backgrounds.

3. Method

Different with the action recognition problem which assumes that motions are conducted by a single human, the sources of the motions and the scenarios where the motion is happening are contributing to DT recognition. To this end, we propose SMTCNN by formulating two sessions - a motion net to capture the temporal motion information, and a source net to capture the spatial information of the source in the motion.

3.1. Motion net

To explore the continuous motions between DT frames, we use the optical flow color maps as the input of the motion net. Let us consider I_1 and $I_2\left(I_1, I_2 : \left(\Omega \to \mathbb{R}^2\right)\right)$ as a pair of frames, we conserve the RGB information of the frames. Using $\overleftarrow{x} = (x, y)$ to denote the intensity of a pixel (x, y) of image space Ω, the optical flow is represented as $\overleftarrow{w} = (\mu, \nu)$, and w reflects the mapping from Ω to $\mathbb{R}^2$. The constraint equation is formulated by considering the constraints in color, gradient and smoothness spaces, and is defined as follows:

$$\begin{cases} E_c(\overleftarrow{w}) = \int_\Omega \Psi(I_2(\overleftarrow{x} + w(\overleftarrow{x})) - I_1(\overleftarrow{x}))d\overleftarrow{x}, \\ E_g(\overleftarrow{w}) = \int_\Omega \Psi(\nabla I_2(\overleftarrow{x} + w(\overleftarrow{x})) - \nabla I_1(\overleftarrow{x}))d\overleftarrow{x}, \\ E_s(\overleftarrow{w}) = \int_\Omega \Psi\left(\nabla\mu(\overleftarrow{x})^2 + \nabla\nu(\overleftarrow{x})^2\right)d\overleftarrow{x}. \end{cases} \tag{1}$$

Here, we define $\Psi(s^2)$ as $\sqrt{s^2 + 10^{-6}}$, and ∇ denotes the orientation of gradients. By constructing the Euler–Lagrange equation and followed by Gauss–Seidel iterations, we transform the estimated optical flow to color maps according to [41].

We further use just noticeable difference (JND), which refers to the threshold above which a distortion becomes visible to the majority of observers, to improve the performance of our optical flow color maps. The computation of JND depends on two components - luminance adaptation (LA) and contrast masking (CM). LA refers to the masking effect towards background luminance, and is computed following [42] as:

$$LA(x,y) = \begin{cases} 17 \times \left(1 - \sqrt{B(x,y)/127}\right) + 3, & \textit{if } B(x,y) \le 127, \\ 3 \times (B(x,y) - 127)/128 + 3, & \textit{otherwise}, \end{cases} \tag{2}$$

where $B(x,y)$ denotes the brightness of the optical flow color maps. Contrast masking indicates the visibility reduction of one visual signal at the presence of another one, which is often composed of two components - edge masking (EM) and texture masking (TM). We first decompose the optical flow color map M into two parts ($M = U + V$), where U denotes the structure optical flow color map while V denotes the texture optical flow color map. EM is estimated from U while TM is estimated from V. For $M(x,y)$, $CM(x,y)$ is defined as $EM^\mu(x,y) + TM^\nu(x,y)$, and the

$EM^{\mu}(x,y)$ is relevant to a spatial contrast with respective to μ, weighted by a constant β and a weight W_e. $TM^{\nu}(x,y)$ is computed using the spatial contrast for ν, also weighted by β and a distinguishing weight W_t. The two weights are set as 1 and 3, respectively and β is defined as 0.117. In our implementation, the spatial contrast is the maximum luminance difference in a 5×5 neighborhood. JND is finally estimated by integrating LA and CM at every pixel, also by considering the minimum value of LA and CM, weighted by 0.3, which formulates a nonlinear additively model according to [42]. For our optical flow color maps, we filter them as:

$$F(x,y) = M(x,y) - JND(x,y), \tag{3}$$

where $F(x,y)$ represents the filtered value.

Given a sequence of $F(x,y)$ sized by $n \times c \times h \times w$, where n denotes the sequence length, c denotes the channel size, h and w denote the height and width of the map. By forwarding the color maps to the pre-trained ResNet [16] until obtaining the very deep features extracted from the *Pool*5 layer, we first expand each deep feature map to a vector x_n^m: For each frame, suppose its feature maps are denoted by $F_{c\times h\times w}$, where the $i^{th}, i \in (1,c)$ feature map is $F_{i\times h\times w} = \left[f_{11}^i, \cdots, f_{1w}^i; f_{21}^i, \cdots, f_{2w}^i; \cdots, f_{h1}^i, \cdots, f_{hw}^i\right]$, our expansion implementation transforms $F_{i\times h\times w}$ to a vector $x_{h\times w}^i = \left(f_{11}^i, \cdots, f_{1w}^i, f_{21}^i, \cdots, f_{2w}^i, \cdots, f_{h1}^i, \cdots, f_{hw}^i\right)$. Considering each frame is deep learnt by c filters, we get a group of deep features $X_1 = \{x_{h\times w}^1, \cdots, x_{h\times w}^m\}$. For the sequence with n frames, to handle all the "words" $\chi = (X_1, \cdots, X_n)$ with a fixed length (the number of filters, i.e. m) for the sequence representation, we concatenate each "alphabet" within each word, to construct a "text". Thus, we get m "sentences" $\{\chi^1, \cdots, \chi^m\}$ to depict the DT motion text $\chi = (\chi_1, \cdots, \chi_n)$. Since all the deep features of each map are temporally considered, the DT motion sequences are spatio-temporally represented.

After spatio-temporal representation, we continue to learn the text by 1D convolution to get its "abstract" - a fully-connected vector for the final classification. Since we have m sentences, the output of each 1D convolution layer is expressed as:

$$x_k^l = \sum_{j=1}^{m_{l-1}} Cov1D\left(w_{jk}^{l-1}, s_j^{l-1}\right) + b_k^l, \tag{4}$$

where x_k^l is the input, b_k^l is the scalar bias of the kth neuron at the l layer, s_j^{l-1} is the output of the kth neuron at the $l-1$ layer, w_{jk}^{l-1} denotes the kernels and the intermediate output of the neuron y_k^l can be written as $y_k^l = f(x_k^l)$, while $s_k^l = y_k^l \downarrow \kappa$, where $\downarrow$ denotes the down-sampling with κ. As the implementations only cover expansion and concatenation, the parameters on the SMTCNN "words" and "text" layers are consistent with the corresponding ones on the *Pool*5 layer, i.e., $s_j^1 = \chi$.

To further learn the SMTCNN "text", the designed 1D convolution module includes 4 convolutional layers, 2 max-pooling layers and 2 fully-connected layers, as detailed in Fig. 2. The filters and the related activation functions are shown in Table 1, and we set the stride as 1 and the paddings as 0. As to the BP error at the transfer layer between 2D convolution module (base architecture-the *Pool*5 layer of ResNet) and 1D convolution module, for χ, suppose the target and output vectors are t^p and $[y_1^L, \cdots, y_N^L]$, respectively. Let E_p denotes the MSE, for the transfer from 2D convolution to 1D convolution, the weight and bias sensitivities

Table 1
Filters and activation functions of our 1D convolution module.

Layer	Filter size	Filter No.	Activation
Conv1	20	32	ReLU
Pool1	2	–	–
Conv2	20	64	ReLU
Pool2	2	–	–
Conv3	20	128	ReLU
Conv4	20	256	ReLU
Dropout(0.3)	–	–	–
Flatten	–	–	–
Fully1	–	1024	ReLU
Dropout(0.2)	–	–	–
Fully2	–	100	ReLU

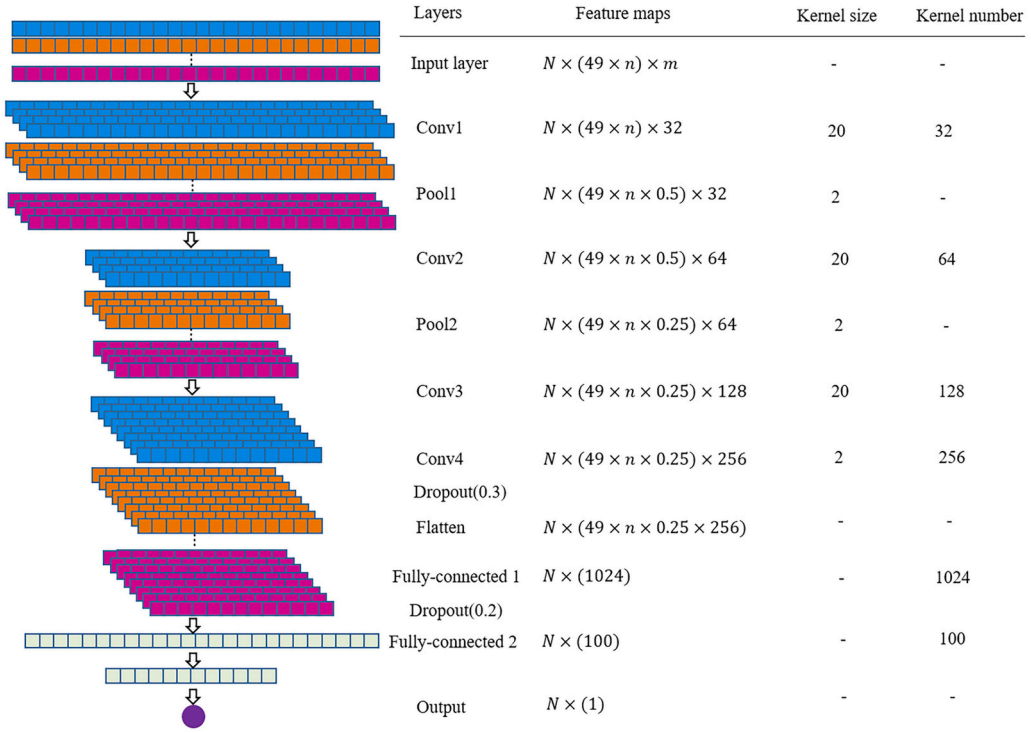


Fig. 2. Architecture of the designed 1D convolution module.

can be expressed as:

$$\frac{\partial E_p}{\partial w_{jk}^1} = conv1D\left(\chi, \Delta_j^2\right), \tag{5}$$

$$\frac{\partial E_p}{\partial b_k^1} = \sum_n \Delta_k^1(n), \tag{6}$$

where

$$\Delta_k^1 = \frac{\partial E_p}{\partial y_k^1}\frac{\partial y_k^1}{\partial x_k^1} = up(\Delta\chi)\kappa^{-1}f'\left(x_k^1\right), \tag{7}$$

where $f()$ denotes the activation function and $up()$ denotes the up-sampling function.

3.2. Source net

To handle the DT videos for input, also to augment our dataset, we cut each video into clips by spatio-temporal sampling. For each video sample $V \in \mathbb{R}^{p_x \times p_y \times p_t}$, it is first temporally sampled via cropping using multiple scales (lengths), i.e., $V_T \in \mathbb{R}^{p_x \times p_y \times q_t}$, where we define $q_t = \{15, 20, 25, 30, 35, 40, 45, 50\}$, and the starting frame is randomly selected in the range of $(0, 15)$. Then V_T is spatially segmented into patches $V_P \in \mathbb{R}^{q_x \times q_y \times q_t}$, with the same imaging center, i.e., each sequence of patches sized by $q_x \times q_y$ is well aligned. Note that the imaging center is obtained on the q_1 frame by cropping with a 224×224 size. As the background of DT videos is relatively stationary, the DT features are primarily reflected by the changing sections, we consider the frame difference distance as the metric to extract the candidate patches. For V_P in the 3D cube sized by $q_x \times q_y \times q_t$, we define a global frame difference descriptor as follows:

$$G_t\left(q_x, q_y, q_t\right) = \sum_{\tau=0}^{q_t-1} D\left(q_x, q_y, \tau\right), \tag{8}$$

where τ denotes the length of the sequence, and we compute

$$D\left(q_x, q_y, t\right) = I\left(q_x, q_y, t\right) - I\left(q_x, q_y, t-1\right), \tag{9}$$

where $t \in [2, \tau]$, and $I\left(q_x, q_y, :\right)$ denotes the brightness of the image.

We further process $G_t\left(q_x, q_y, q_t\right)$ by min-max normalization to highlight its kinematic features, followed by being approximated by $F_t\left(q_x, q_y, q_t\right) = \sum_{i=1}^{q_x}\sum_{j=1}^{q_y} G_t(i, j, q_t)$. The spaiotemporal clips are finally determined by thresholding $F_t\left(q_x, q_y, q_t\right)$ utilizing ζ_t, where ζ_t is an adjustable constant.The input is finally represented as $V_P \in \mathbb{R}^{q_x \times q_y \times q_1} + G_t\left(q_x, q_y, q_t\right)$. We also employ ResNet pre-trained on ImageNet ILSVRC-2012 as the base to recognize the DT subjects, and transfer learning is utilized from the layer of *Cov*1 to *Pool*5 to generate deep features. Intuitively, we sequentially feed the DT frames into the backbone and extract the features of the last pooling layer, followed by a fully-connected layer, which are finally classified using a Softmax layer. The classification results of the DT sequence are combined by the motion and source net using max-voting. The network is optimized using mini-batch momentum SGD algorithm, and the batch size is selected as 64, with $L2$ normalization. We adopt 0.01 as the learning rate and empirically set the momentum value as 0.9, and after 200 epochs, the accuracy is recorded.

4. Experiments

We evaluate the performance of our approach on three widely adopted DT datasets: UCLA, DynTex and DynTex++. The three datasets are varied on volume, size and scene to comprehensively test our approach. The results and comparisons with the state of the art are given in this section.

4.1. Datasets

The UCLA database contains 200 DT videos categorized into 50 classes, with a dimension of $110 \times 160 \times 75$, following the leave-one-out (50-LOO) protocol by default. Derived from the original setup, another three popular sub-datasets are often used as benchmarks: 4 cross-fold validation (4-fold), 9-class breakdown and 8-class breakdown. The 4-fold protocol takes one fourth of each class for testing, while the rest for training, as reported in [13,23]. 9-class breakdown and 8-class breakdown protocols respectively realign the 50 classes into 8 and 9 classes, and one half of the samples are randomly selected for training, while the other half for testing.

DynTex dataset includes over 650 videos, sized by $352 \times 288 \times 250$. Following [14], 679 DT videos are employed for training and testing utilizing LOO scheme in our experiment. Moreover, we adopt 4 challenging subsets of DynTex for evaluation. DynTex35 consists of 10 classes founded by splitting from 35 videos, each of which is treated as a category to form 8 non-overlapping sub-DTs via clipping points along X, Y, T axes randomly and separately, except that at the half of them. DynTex-alpha contains 60 DT videos equally categorized into "grass", "sea" and "trees". DynTex-beta contains 162 DT videos categorized into 10 classes, and Dyntex-gamma contains 264 DT videos categorized into 10 classes.

DynTex++ [43] dataset consists of 36 classes with 100 DT sequences for each of them, deriving from DynTex by splitting 345 raw videos with a fixed size $50 \times 50 \times 50$. One half of the samples from each class is randomly selected for training while the rest for testing, following the default protocol.

4.2. Results

We record the training and testing loss to show the evolutions of our network, as shown in Fig. 3(a) and Fig. 3(c), also, the accuracy evolutions in training and testing every 10 epochs are given in Fig. 3(b) and Fig. 3(d). From Fig. 3, we can find out that the convergence of SMTCNN happens during 50–100 epochs on UCLA 8-class, UCLA 9-class, DynTex35, DynTex-alpha, DynTex-beta, DynTex-gamma and DynTex++. For UCLA 50-LOO and UCLA 4-fold, it converges very fast within 50 epochs, and until 180 epochs, SMTCNN is very close to the final convergence. Among all the datasets, SMTCNN converges fastest on DynTex++. The loss on both training and testing is small, and the accuracy on both training and testing is high, with small difference, indicating the good generalization ability of SMTCNN. Thus, we can conclude that the distinctions of convergence of SMTCNN among different datasets are related to the volume size, category and training/testing configurations. Moreover, we compare our method with a range of state-of-the-art approaches [13,20–31,44,45] on three DT datasets (UCLA, DynTex and DynTex++), the results are reported in Table 2. Consistent high accuracy on the benchmarks are obtained using the proposed SMTCNN, which indicates its good performance. Moreover, we compare our method with a range of state-of-the-art approaches [13,20–31,44,45] on several recent DT datasets (UCLA, DynTex and DynTex++), the results are reported in Table 2. Consistent high accuracy on the benchmarks are obtained using the proposed SMTCNN, which indicates its good performance.

UCLA: The recognition rates on 50-LOO and 50–4-fold are higher than the 9-class breakdown and 8-class breakdown protocols using our and other algorithms. Despite the local feature based methods (MEWLSP [21] and LBP-TOP [20]) dominates the early DT recognition research, methods considering the DT characteristics (MLDP-TOP [23,45]) and learning based methods (Orthogonal Tensor DL [26] and Equiangular Kernel DL [27]) perform better nowadays, Plus, since the volume of UCLA is small (200 videos categorized into 50 classes), deep learning

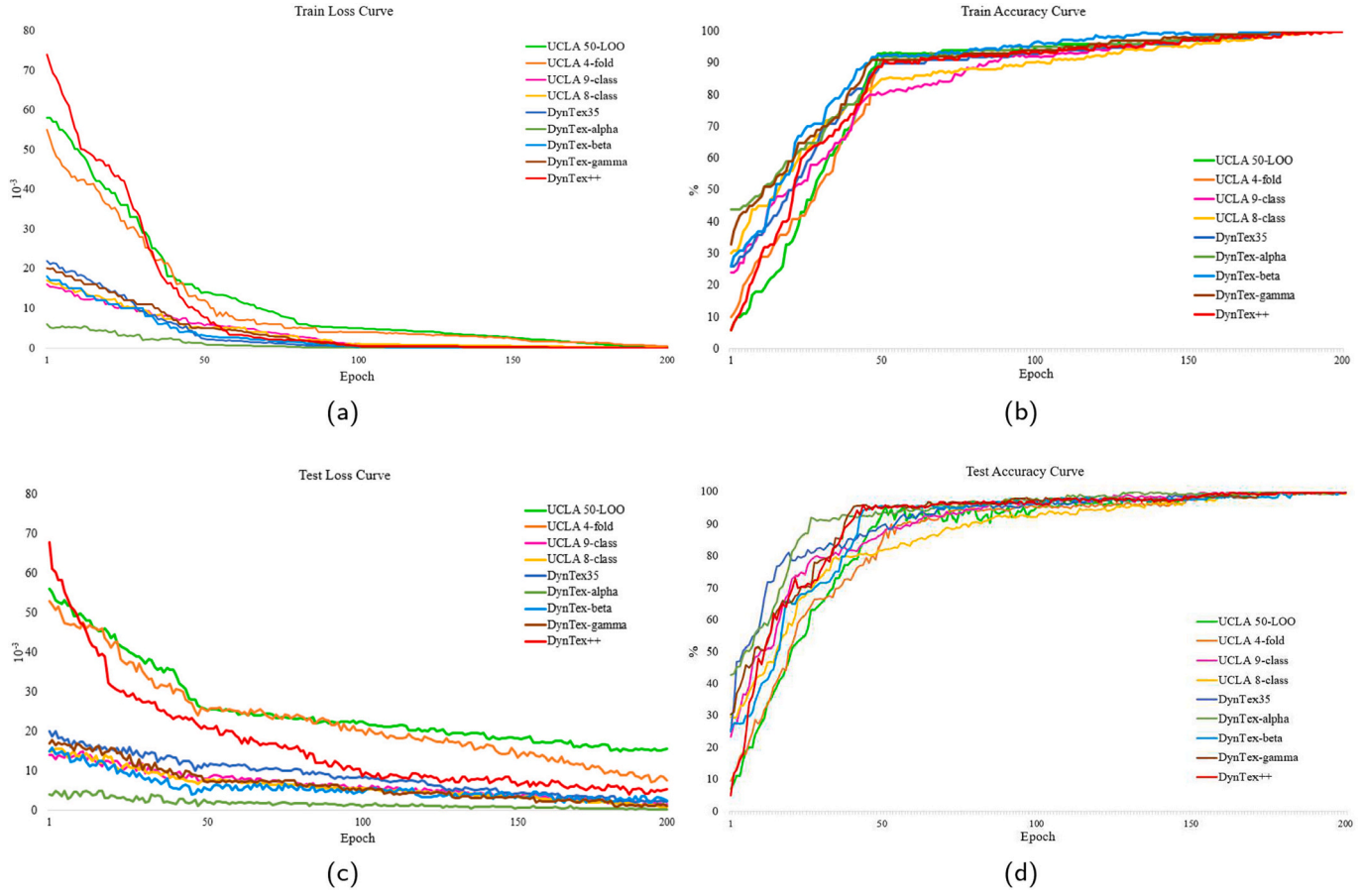


Fig. 3. Training (a and b) and testing (c and d) evolutions of SMTCNN on benchmarks.

Table 2
Comparison of recognition rates (%) of different methods on benchmark DT datasets.

Methods	UCLA 50-LOO	UCLA 4-fold	UCLA 9-class	UCLA 8-class	DynTex35	DynTex-alpha	DynTex-beta	DynTex-gamma	DynTex++
Orthogonal Tensor DL [26]	–	99.80	98.20	99.50	–	87.80	76.70	74.80	94.70
Equiangular Kernel DL [27]	–	–	–	–	–	88.80	77.40	75.60	93.40
MEWLSP [21]	96.50	96.50	98.55	98.04	99.71	–	–	–	98.48
LBP-TOP [20]	–	94.50	96.00	93.67	92.45	98.33	88.89	84.85	94.05
STLS [25]	–	99.50	97.40	99.50	98.20	89.40	80.80	79.80	94.50
CVLBC [22]	98.50	99.00	99.20	99.02	98.86	–	–	–	91.31
MMDP-B [23]	99.50	98.50	98.05	97.61	98.86	96.67	88.27	93.18	95.82
MLDP-TOP [23]	99.00	99.00	98.60	97.72	98.29	95.00	91.98	91.29	95.50
CLSP-TOP [45]	97.00	97.00	96.50	96.09	98.86	96.67	88.89	92.80	94.02
FDT [24]	98.50	99.00	97.70	99.35	98.86	98.33	93.21	91.67	95.31
FD-MAP [24]	99.50	99.00	99.35	99.57	98.86	98.33	92.59	91.67	95.69
PCANet-TOP [44]	99.50	–	–	–	–	96.67	90.74	89.39	–
DNG [29]	–	–	99.60	99.40	–	–	–	–	93.80
DFS [28]	–	100	97.5	99.20	97.16	85.24	76.93	74.82	91.70
st-TCoF [30]	–	–	–	–	–	100	100	98.11	–
D3 [31]	–	–	–	–	–	100	100	98.11	–
DT-AlexNet [13]	–	99.50	98.05	98.48	–	100	99.38	99.62	98.18
DT-GoogleNet [13]	–	99.50	98.35	99.02	–	100	100	99.62	98.58
SMTCNN	100	100	99.62	99.36	100	100	100	99.75	99.62

methods [13,30,31,44] are limited to inadequate network training and do not perform better than hand-crafted feature based methods [20–29,45]. Optical flow based methods-FDT [24] and FD-MAP [24] obtain very high recognition rates on UCLA, reflecting the importance of dynamic features for DT recognition. Also, it is easy to find out that LBP based approaches [20,21,45] are rather active in recent years, the high-level accuracy suggest the robustness of the classic hand-crafted feature. Benefiting from our spatial cropping scheme, which strengthens the data augmentation process, SMTCNN outperforms state-of-the-art deep learning approaches, and in contrast to the recent hand-crafted feature based methods, the accuracy is (100% for UCLA 50-LOO, 100% for 50–4-fold and 99.62% for 9-class) or very close to (99.36% for 8-class) the best (99.75% by FD-MAP [24]). For UCLA 9-class and 8-class, the accuracy has not reached 100% yet, Fig. 4(a) and Fig. 4(b) show the confusion matrix using our approach, from which we can see that classification for classes “sea”, “smoke” and “water” are more

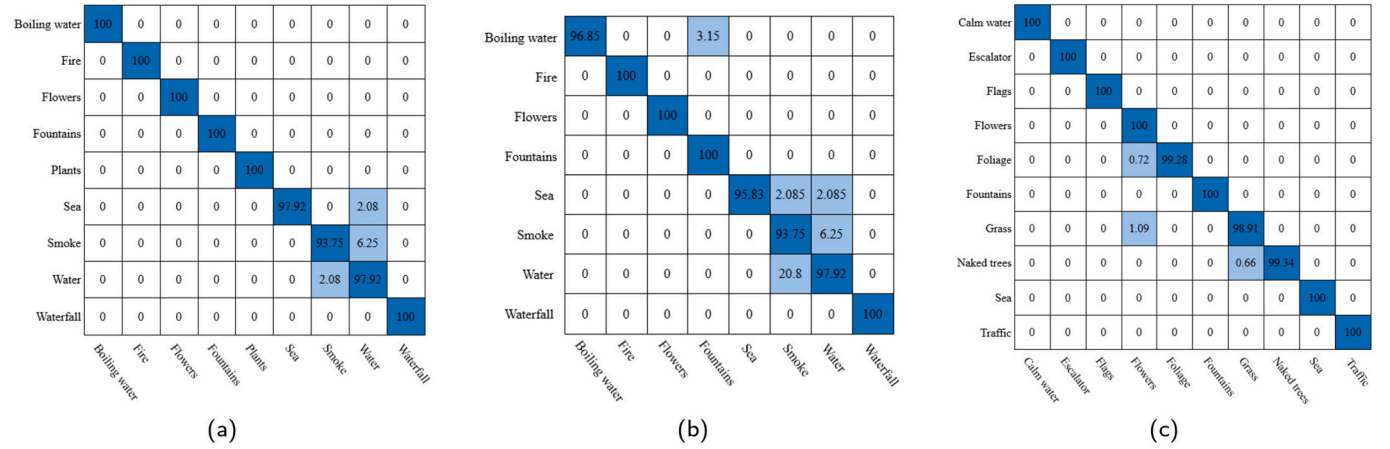


Fig. 4. Confusion matrix using SMTCNN on UCLA 9-class (a), UCLA 8-class and DynTex-gamma (c).

challenging maybe due to they are more similar on appearance.

DynTex: In all, deep learning methods are superior to hand-crafted feature based methods on DynTex owing to the advantage of CNNs as well as the relatively big volume of this benchmark. The accuracy of hand-crafted feature based methods range from 74.80% to 99.75% on DynTex-gamma, and by rearranging the raw videos to DynTex++, the recognition results are greatly boosted. Optical flow based methods-FDT [24] and FD-MAP [24] perform best among hand-crafted feature based methods, showing the significance of temporal features of DT. Also, the results using conventional local features such as MEWLSP [21] are promising, proving that the popular scheme focusing from local to global is applicable for the DT feature extraction. As well, unlike Orthogonal Tensor DL [26], Equiangular Kernel DL [27] and DFS [28], which get good results on UCLA and low recognition rates on DynTex, SMTCNN reaches a very high accuracy on the five derived sub-datasets, especially on the DynTex35, DynTex-alpha and DynTex-beta, and the impact of varied datasets is small, which shows great preponderance compared to hand-crafted ones. Actually, the results using DT-AlexNet [13] and DT-GoogleNet [13] are comparable to ours, indicating not only the changing information plays an important role, but also the regularity of DT videos is significant for the recognition. For DynTex35, DynTex-alpha and DynTex-beta, our method all gets 100% recognition rate, the accuracy on DynTex-gamma is 99.75%, and the confusion matrix is given in Fig. 4(c), from which we can see that classes “foliage”, “grass” and “naked trees” have not been perfectly recognized yet.

4.3. Ablation study

4.3.1. Effectiveness of sub-nets

SMTCNN is fused by two sub-nets - motion net to capture temporal dynamic features and source net to capture spatial salient features. To verify their contributions to SMTCNN, we give the results using the two sub-nets separately on UCLA, DynTex and DynTex++ in Table 3. Results using motion net outperform source net to a big extent, which aligns with the statement proved by [6] that the temporal features provide more discriminant information than spatial features. By fusing the two sub-nets, our SMTCNN obtains more promising recognition rates on the benchmarks, which indicates that the two sub-nets are complementary.

4.3.2. Effectiveness of backbones

The base architecture for SMTCNN is selected as ResNet considering its strong classification ability, to test the feasibility of other CNNs and compare their performance in DT recognition task, we replace ResNet utilizing AlexNet [46] and GoogleNet [47] with the identical post-processing settings, respectively. The recognition rates are shown in Table 4. Results illustrate that the feature learning abilities of ResNet is a little better than AlexNet and GoogleNet, which is broadly verified. Nevertheless, the overall gap using different architectures within our framework is small, suggesting that the robustness of our approach is strong.

4.3.3. Effectiveness of JND

We employ JND to pre-process the optical flow color maps, in order to let SMTCNN learn the temporal DT information from human’s perspective. In spite of JND is popular in the field of graphics, as far as we know, this is the first work to adopt JND to deal with the DT recognition problem. To demonstrate the effectiveness of JND, we also directly input the optical flow color maps to our motion net without JND, Table 5 shows the results and comparisons on benchmarks. We can find out that JND can enhance the accuracy by about 2.3% on UCLA, about 1.2% on DynTex and about 1.1% on DynTex++. We wish our results can inspire more relevant joint work of JND and visual classification.

5. Conclusion

This paper presents a spatio-temporal semantic deep learning framework - SMTCNN for 3D DT recognition. SMTCNN is concerned with DT’ motions as well as the sources of the motions and the scenarios where the motion is happening in DT sequences, by formulating a source net with a DT spatio-temporal segments as input using an arbitrary backbone and a motion net dependent on a novel module containing expansion and concatenation implementations on 2D CNN tensors, followed by a new 1D CNN architecture. A series of experimental results demonstrate our novel framework and the effectiveness of the designed modules. In spite of only DT recognition problem is considered in this paper, the proposed SMTCNN is applicable for other visual classification tasks and real-word applications, which we would like to explore in our

Table 3
Comparisons of recognition rates (%) of sub-nets.

Neural networks	UCLA 50-LOO	UCLA 4-fold	UCLA 9-class	UCLA 8-class	DynTex35	DynTex-alpha	DynTex-beta	DynTex-gamma	DynTex++
Motion net	98.17	98.00	97.58	97.75	99.24	97.79	99.00	98.25	99.00
Source net	95.58	96.26	95.27	94.06	95.78	96.02	97.13	96.44	97.68
Fusion-SMTCNN	100	100	99.62	99.36	100	100	100	99.75	99.62

Table 4
Comparisons of recognition rates (%) using different backbones.

Base architecture	UCLA 50-LOO	UCLA 4-fold	UCLA 9-class	UCLA 8-class	DynTex35	DynTex-alpha	DynTex-beta	DynTex-gamma	DynTex++
SMTCNN-AlexNet	98.89	99.00	99.70	98.04	98.58	99.75	100	98.77	99.31
SMTCNN-GoogleNet	99.02	99.35	99.48	98.65	100	99.75	100	99.00	99.45
SMTCNN-ResNet	100	100	99.62	99.36	100	100	100	99.75	99.62

Table 5
Comparisons of recognition rates (%) with/without JND.

Scheme	UCLA 50-LOO	UCLA 4-fold	UCLA 9-class	UCLA 8-class	DynTex35	DynTex-alpha	DynTex-beta	DynTex-gamma	DynTex++
Without JND	99.02	98.86	96.19	96.02	98.78	99.23	99.50	97.79	98.56
With JND-SMTCNN	100	100	99.62	99.36	100	100	100	99.75	99.62

future research.

CRediT authorship contribution statement

Liangliang Wang: Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Lei Zhou:** Writing – review & editing, Methodology. **Peidong Liang:** Supervision, Software, Resources, Data curation. **Ke Wang:** Software, Investigation. **Lianzheng Ge:** Validation, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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